Problem 2 | Problems T5_MechanicalProperties.pdf #2 | Tensile test · true stress · area reduction

Tensile Test of a 15 mm Cylinder: Strength, Resilience, True Stress and Area Reduction

Original statement · Problems T5_MechanicalProperties.pdf

2. A cylindrical sample made from a metallic material (assume a Poisson ratio of ν = 0.25) of 15 mm in diameter and a calibrated length of 50 mm was subjected to a tensile test, and the results are shown in the table below:

Point	Force (kN)	ΔL (mm)
1	70	0.25
2	120	0.40
3	150	0.50
4	160	0.60
5	170	0.75
6	200	1.75
7	220	3.00
8	233	5.00
9	233	6.50
10	220	8.00

The diameter of the sample after fracture was 12.45 mm. Calculate:

- *a. The tensile strength.*
- *b. The Young's Modulus (or elastic modulus).*
- *c. The yield stress.*
- *d. The modulus of resilience.*
- *e. The real tension for an engineering deformation of 0.02.*
- *f. The percentage of deformation to fracture and percentage of area reduction.*

(The sheet prints the table in two rows; here each point has its own row.)

Official Solution: a. $\sigma_{UTS} = 1319$ **MPa; b.** $E = 85$ **GPa; c.** $\sigma_y = 920$ **MPa; d.** $U_r = 4.98$ **MPa; e.** $\sigma_r = 1030$ **MPa; f. 15 %, 7.3 %**

HYPOTHESES & PARAMETERS

Hypotheses

1. **Uniaxial, uniform stress.** Up to the necking the stress is uniform along the gauge length, so the engineering quantities use the original area and length [Session 11 Slide 9].
2. **Linear elastic start.** Hooke's law $\sigma = E\varepsilon$ holds on the first points of the curve [Session 11 Slide 13].
3. **Yield strength by the 0.2 % offset.** The statement does not name a criterion; the conventional one for metals is the 0.2 % offset [Session 11 Slide 28].
4. **Straight segments between table rows**, so readings between two rows are linear interpolations.
5. **Uniform deformation before the necking.** The volume is constant, $A\,l = A_0 l_0$, until the maximum load, where the necking starts [Session 11 Slides 30 and 38]. The true stress of item e is computed in that range.
6. **Elastic recovery after the fracture.** The elastic strain disappears when the load does, and what is left is the plastic strain [Session 11 Slide 32]. The reduction of area uses the measured final diameter [Session 11 Slide 33].
7. **The Poisson ratio is not needed.** $\nu = 0.25$ is a datum of the statement, but none of the six items uses it: the only lateral information is the final diameter, which is measured.

Data

Quantity	Symbol	Value	In SI
Diameter	d_0	15 mm	15×10^{-3} m
Gauge length	l_0	50 mm	50×10^{-3} m
Diameter after fracture	d_f	12.45 mm	12.45×10^{-3} m
Poisson ratio (not used)	ν	0.25	
Load, elongation	F, ΔL	table	kN, mm

Unknowns: σ_{UTS} (a), E (b), σ_y (c), U_r (d), σ_r at $\varepsilon_n = 0.02$ (e), the elongation at fracture and the reduction of area (f).

MODEL & STRATEGY

The only direction is the axis of the cylinder and the stress is uniaxial, so there is no change of basis.

Strategy. As in any tensile test, the first step is to turn the table into engineering stress and strain,

$$\sigma_n = \frac{F}{A_0} \qquad \varepsilon_n = \frac{\Delta L}{l_0}$$

and then read each item from it: the maximum stress (a), the slope of the straight part by least squares through the origin (b), the offset construction solved on the crossing segment (c), $U_r = \sigma_y^2 / 2E$ (d) [Session 11 Slide 35].

True stress (e). With constant volume, $A\,l = A_0 l_0$, so $A = A_0 l_0 / l = A_0/(1+\varepsilon_n)$. The true stress is then

$$\sigma_r = \frac{F}{A} = \frac{F}{A_0}\,(1+\varepsilon_n) = \sigma_n\,(1+\varepsilon_n)$$

which is the relation of the slide for the uniform deformation [Session 11 Slide 38].

STEP-BY-STEP CALCULATION

1. Cross-section

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi\,(15\ \text{mm})^2}{4}$$

The square is $15^2 = 225$, then $\pi \times 225 = 706.86$ and dividing by 4:

$$A_0 = 176.71\ \text{mm}^2$$

2. Engineering stress and strain of every point

The loads are in kN, so $F = 1000 \times F_{kN}$ newtons. For example, point 1 is $F = 70\,000$ N and $\Delta L = 0.25$ mm, so $\sigma_n = 70\,000/176.71 = 396.1$ MPa and $\varepsilon_n = 0.25/50 = 0.0050$. All the points:

Point	F (N)	ΔL (mm)	σ_n (MPa)	ε_n	σ_n/ε_n (GPa)
1	70 000	0.25	396.1	0.0050	79.2
2	120 000	0.40	679.1	0.0080	84.9
3	150 000	0.50	848.8	0.0100	84.9
4	160 000	0.60	905.4	0.0120	75.5
5	170 000	0.75	962.0	0.0150	64.1
6	200 000	1.75	1131.8	0.0350	32.3
7	220 000	3.00	1244.9	0.0600	20.7
8	233 000	5.00	1318.5	0.1000	13.2
9	233 000	6.50	1318.5	0.1300	10.1
10	220 000	8.00	1244.9	0.1600	7.8

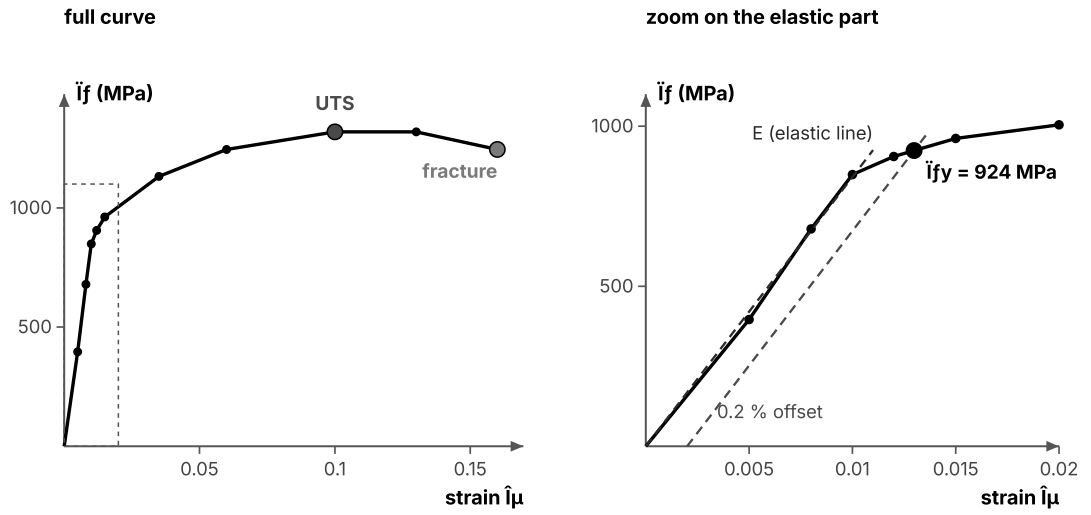


Figure 2. Problem 2: engineering stress-strain curve built from the data of the sheet (every dot is a table row), with the elastic line and the 0.2 % offset construction.

a) Tensile strength

The maximum load is $F_{max} = 233\,000$ N, reached at points 8 and 9 ($\varepsilon_n = 0.100$ and 0.130: the load stays at its maximum while the sample elongates). The area A_0 is constant in the engineering stress, so

$$\sigma_{UTS} = \frac{F_{max}}{A_0} = \frac{233\,000\ \text{N}}{176.71\ \text{mm}^2} = \mathbf{1318.5\ MPa}$$

b) Young's modulus

The secant modulus is 79 to 85 GPa for points 1 to 3, and it drops to 75 GPa at point 4 and to 64 GPa at point 5: the linear zone ends after point 3. The fit through the origin with points 1 to 3 is:

Point	ε_i	σ_i (MPa)	$\sigma_i \varepsilon_i$ (MPa)	ε_i^2
1	0.0050	396.1	1.9806	2.500×10^{-5}
2	0.0080	679.1	5.4325	6.400×10^{-5}
3	0.0100	848.8	8.4883	1.000×10^{-4}

$$E = \frac{\sum \sigma_i \varepsilon_i}{\sum \varepsilon_i^2} = \frac{15.901}{1.8900 \times 10^{-4}} = 84134\ \text{MPa} = \mathbf{84.1\ GPa}$$

c) Yield stress (0.2 % offset)

The offset line is $\sigma_{off} = E(\varepsilon - 0.002)$ and the sign of $g = \sigma_n - E(\varepsilon_n - 0.002)$ tells where the curve crosses it. The sign changes between points 4 and 5:

Point	ε_n	σ_n (MPa)	$E(\varepsilon_n - 0.002)$ (MPa)	$g = \sigma_n - E(\varepsilon_n - 0.002)$ (MPa)
3	0.01000	848.8	673.1	175.8
4	0.01200	905.4	841.3	64.1
5	0.01500	962.0	1093.7	-131.7
6	0.03500	1131.8	2776.4	-1644.7

On that segment

$$m = \frac{962.00 - 905.41}{0.01500 - 0.01200} = 18862.8 \text{ MPa} \qquad \varepsilon^* = \frac{\sigma_4 - m\,\varepsilon_4 + 0.002\,E}{E - m} = \frac{905.41 - 18862.8 \times 0.01200 + 0.002 \times 84134}{84134 - 18862.8} = \frac{847.329}{65271.3} = 0.0129817$$

$$\sigma_y = E\,(\varepsilon^* - 0.002) = 84134 \times (0.0129817 - 0.002) = \mathbf{923.9\ MPa}$$

d) Modulus of resilience

$$U_r = \frac{\sigma_y^2}{2E} = \frac{(923.9\ \text{MPa})^2}{2 \times 84134\ \text{MPa}} = \mathbf{5.07\ MPa}$$

($1\ \text{MPa} = 1\ \text{MJ/m}^3$.)

e) True stress for $\varepsilon_n = 0.02$

The elongation that corresponds to $\varepsilon_n = 0.02$ is $\Delta L = 0.02 \times 50 = 1.00$ mm, which is not a row of the table. It is between points 5 ($\Delta L = 0.75$ mm, $F = 170$ kN) and 6 ($\Delta L = 1.75$ mm, $F = 200$ kN):

$$F = F_5 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(F_6 - F_5) = 170 + \frac{1.00 - 0.75}{1.75 - 0.75}\,(200 - 170) = 177.5\ \text{kN}$$

$$\sigma_n = \frac{177\,500\ \text{N}}{176.71\ \text{mm}^2} = 1004.4\ \text{MPa}$$

This strain is below the one of the maximum load ($\varepsilon_n = 0.100$), so the deformation is still uniform and the relation of section 2 applies:

$$\sigma_r = \sigma_n\,(1+\varepsilon_n) = 1004.4 \times (1+0.02) = \mathbf{1024.5\ MPa}$$

f) Elongation at fracture and reduction of area

The last row of the table is the fracture: $\Delta L = 8.00$ mm and $F = 220\,000$ N, so

$$\varepsilon_f = \frac{8.00}{50} = 0.160 \qquad \sigma_f = \frac{220\,000}{176.71} = 1244.9\ \text{MPa}$$

The elastic part of that strain is recovered when the sample breaks, $\varepsilon_e = \sigma_f / E$, and what is left is the plastic strain [Session 11 Slide 32]:

$$\varepsilon_e = \frac{1244.9}{84134} = 0.01480 \qquad \varepsilon_p = 0.160 - 0.01480 = 0.1452 \qquad EL = \mathbf{14.5\ \%}$$

For the reduction of area, the final area from the final diameter is

$$A_f = \frac{\pi\,(12.45\ \text{mm})^2}{4} = \frac{\pi \times 155.0025}{4} = 121.74\ \text{mm}^2$$

$$RA\,(\%) = \frac{A_0 - A_f}{A_0} \times 100 = \frac{176.71 - 121.74}{176.71} \times 100 = \mathbf{31.1\ \%}$$

RESULT & CHECKS

Checks

- **Units.** Stresses in N/mm² = MPa, strains without units, E and U_r in MPa.
- **Order.** $\sigma_y = 923.9$ MPa $< \sigma_{UTS} = 1318.5$ MPa, and the stress rises 395 MPa between the yield and the maximum load.

- **True and engineering stress.** The true stress of item e, 1024.5 MPa, is 2 % above the engineering one, 1004.4 MPa, as the factor $1 + \varepsilon_n$ says.
- **Elongation against area reduction.** $RA = 31.1$ % is larger than $EL = 14.5$ %, as expected when a ductile sample necks: the section shrinks much more at the neck than the length grows over the whole gauge length.
- **Neck.** The stress that acts on the neck at the fracture is $F_f / A_f = 220\,000 \text{ N} / 121.74 \text{ mm}^2 = 1807$ MPa, far above the engineering fracture stress of 1244.9 MPa because the area is much smaller than A_0.

Interpretation

The sample reaches 1.3 GPa with a plateau of maximum load, and it fails after a clear neck. The measured final diameter, not Poisson's ratio, gives the lateral information: Poisson's ratio only describes the elastic part of the deformation.

Check against the official solution

Item	Official	Computed	Comment
a. σ_{UTS}	1319 MPa	1318.5 MPa	agrees
b. E	85 GPa	84.1 GPa	agrees (1 %); the key corresponds to the slope between points 2 and 3
c. σ_y	920 MPa	923.9 MPa	agrees (0.5 %)
d. U_r	4.98 MPa	5.07 MPa	agrees (2 %); the key uses $E = 85$ GPa and $\sigma_y = 920$ MPa
e. σ_r	1030 MPa	1024.5 MPa	agrees (0.5 %): the key reads $\sigma_n \approx 1010$ MPa at $\varepsilon_n = 0.02$
f. elongation	15 %	14.5 %	agrees (rounded)
f. area reduction	7.3 %	31.1 %	**does not agree**

> **DISCREPANCY WITH THE OFFICIAL SOLUTION: AREA REDUCTION**
>
> The official 7.3 % cannot be reproduced from the data of the statement. With $d_0 = 15$ mm and $d_f = 12.45$ mm the final area is $A_f = 121.74$ mm² and the reduction is 31.1 %. A reduction of 7.3 % would need a final area of $0.927\,A_0$, that is, a final diameter of $15\sqrt{0.927} = 14.44$ mm. The computed value is kept.